

SVETLOLOBOVO, Russian Federation

WMO: 296640

Lat: 55.0891N

Lon: 90.8485E

Elev: 1067

StdP: 14.14

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.9	-31.0	-41.2	0.5	-35.7	-36.1	0.7	-30.9	25.4	20.0	24.1	19.6	1.5	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	83.5	66.3	80.4	64.7	77.4	63.4	68.6	78.6	66.9	76.7	65.3	74.6	5.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	97.4	72.7	63.5	91.2	71.1	61.8	85.8	69.5	33.6	79.1	32.1	77.2	30.9	75.2	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	19.3	16.5	-38.4	89.0	6.2	3.2	-42.9	91.3	-46.5	93.2	-49.9	95.0	-54.4	97.3		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	-1.8	2.4	20.3	38.0	47.8	62.0	64.5	60.6	47.9	34.7	15.2	7.0
	DBStd	25.69	17.56	15.92	13.23	8.22	8.85	7.04	4.67	5.51	7.35	9.18	14.17	16.03
	HDD50	7365	1607	1332	920	371	152	7	0	1	125	474	1044	1332
	HDD65	11654	2072	1752	1385	810	535	134	68	155	514	938	1494	1797
	CDD50	1305	0	0	0	11	85	366	448	332	62	1	0	0
	CDD65	118	0	0	0	0	3	43	51	20	1	0	0	0
	CDH74	1471	0	0	0	4	61	636	512	234	24	0	0	0
	CDH80	289	0	0	0	0	8	145	96	38	2	0	0	0
	WSAvg	6.1	6.7	5.6	6.2	7.3	8.3	4.7	4.2	4.1	5.1	6.9	7.1	6.9
(15) Precipitation	PrecAvg	14.2	0.3	0.4	0.4	0.7	1.1	2.4	2.6	2.6	1.5	0.9	0.7	0.6
	PrecMax	17.8	0.7	1.2	0.8	1.8	2.4	4.7	5.3	7.7	3.5	2.0	1.5	1.4
	PrecMin	10.0	0.1	0.0	0.1	0.1	0.3	1.2	0.7	0.7	0.5	0.3	0.2	0.1
	PrecStd	2.2	0.2	0.3	0.2	0.4	0.5	0.8	1.2	1.4	0.7	0.4	0.3	0.4
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	35.8	51.8	69.5	80.0	87.8	86.5	84.1	77.6	61.7	48.1	36.0
		MCWB	29.5	33.2	41.7	53.2	58.1	66.0	67.3	69.7	62.0	51.3	43.4	31.6
	2%	DB	29.6	32.2	45.8	62.4	74.5	84.2	82.9	79.8	70.2	56.1	41.3	32.4
		MCWB	26.4	28.9	38.1	48.3	55.6	65.4	66.4	66.5	59.9	47.6	36.6	29.0
	5%	DB	25.4	29.3	41.7	57.6	69.8	80.8	79.7	76.1	65.2	51.9	36.8	30.1
		MCWB	22.2	26.3	35.8	46.2	53.3	64.1	65.8	64.4	56.9	45.4	32.9	27.3
	10%	DB	22.4	25.9	38.0	53.3	64.8	77.5	76.6	72.2	60.5	48.4	33.1	27.0
		MCWB	19.7	23.4	33.4	43.6	51.9	62.2	64.7	62.6	54.1	42.3	30.1	24.5
	0.4%	WB	30.1	33.3	42.6	53.2	60.5	69.8	70.6	71.0	63.1	51.8	43.2	31.9
		MCDB	33.0	35.4	50.6	68.1	73.7	82.1	79.9	81.5	73.2	59.2	48.2	35.2
	2%	WB	26.2	29.4	38.9	49.5	57.7	67.6	68.6	68.1	60.4	48.8	36.5	29.7
		MCDB	29.9	31.6	44.7	59.5	70.2	78.9	77.9	77.3	68.5	55.2	40.5	32.1
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	22.4	26.6	36.0	46.9	55.1	65.8	67.2	65.6	57.7	45.8	33.3	27.3
		MCDB	25.2	29.4	41.0	56.7	66.3	77.3	76.8	73.1	64.3	51.5	36.6	29.9
	10%	WB	19.6	23.1	33.6	44.1	52.7	64.2	65.7	64.0	54.9	42.9	30.5	24.6
		MCDB	22.3	25.9	37.9	52.9	64.3	75.5	75.1	70.9	59.7	47.8	33.0	27.4
		MDBR	14.5	16.8	19.1	19.9	20.5	22.9	20.0	19.7	18.7	14.7	13.7	14.1
		MCDBR	13.0	14.5	18.9	26.7	31.1	29.5	26.4	26.1	27.7	22.2	12.3	9.9
(36) Mean Daily Temperature Range	5% DB	MCWBR	11.3	12.5	12.9	15.4	16.2	13.2	12.4	13.5	17.0	14.5	9.2	8.7
		MCDBR	12.6	12.9	17.8	25.2	26.1	25.0	22.4	22.4	26.1	21.4	12.2	9.9
	5% WB	MCWBR	11.0	10.9	12.6	15.4	14.8	12.0	11.7	12.5	16.3	14.5	9.7	9.0
(40) Clear-Sky Solar Irradiance	taub		0.277	0.312	0.360	0.401	0.410	0.405	0.427	0.410	0.370	0.347	0.312	0.279
	taud		2.035	2.008	1.946	2.092	2.185	2.301	2.246	2.262	2.314	2.236	2.056	1.974
	Ebn at Noon		196	233	248	258	264	267	257	254	249	221	177	159
	Edh at Noon		22	32	44	44	43	39	40	37	31	26	21	18
(44) All-Sky Solar Radiation	RadAvg		278	598	1058	1416	1602	1891	1674	1379	929	495	245	175
	RadStd		28	47	97	80	182	162	95	124	102	47	30	21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page